REKLUSE® RADIUS™ cx

INSTALLATION GUIDE

Harley-Davidson
Big Twins
Hydraulic Acuated

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OVERVIEW

This kit replaces many of the OE (Original Equipment) or “stock” clutch parts. These parts are designed specifically for your motorcycle to ensure optimal performance. The following is a summary of what is replaced:

- OE clutch pack (frictions and drive plates)
- OE clutch hub, pressure plate, and springs

⚠CAUTION

Do not dyno-test this product without reviewing the included dynamometer document.

This clutch replaces the OE assist and slip clutch. Eliminating the assist and slip clutch provides better modulation and a superior clutch lever feel.

TOOLS

- Hex key set (SAE)
- T27 & T70 Torx bits
- Torque wrench (capable of 170 lb.ft)
- 10 mm, 1/2” & 1 3/16” sockets
- End wrenches (SAE)
- Hydraulic shop press
- Snap ring pliers (internal & external)

INCLUDED PARTS

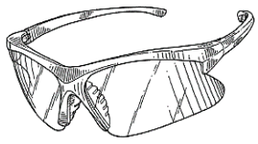
Refer to the included **Parts Fiche** for a detail of the components. Visit www.rekluse.com/support for a full parts fiche illustration and part numbers.

BEFORE YOU BEGIN

Rekluse recommends installing a new basket bearing with this kit. Harley Davidson bearing part numbers:

- **M8:37000168**
- **2011-2017 Twin Cam:37906-11**
- **2007-2010 Twin Cam: 37906-90**

INSTALLATION TIPS



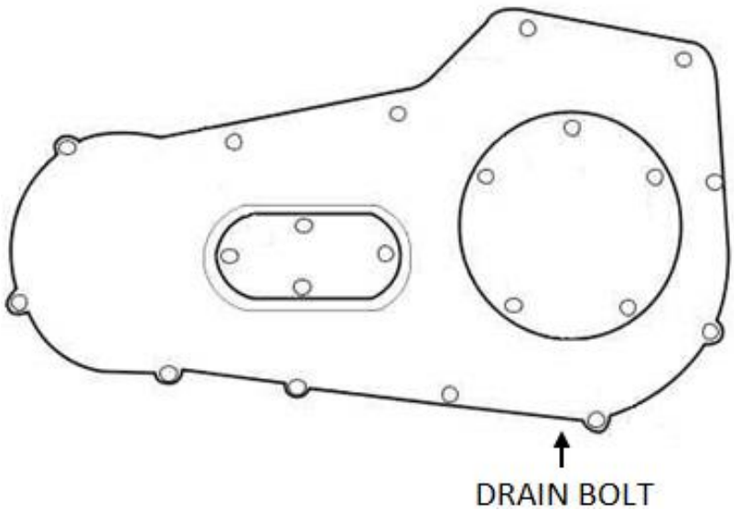


- Read the separate included Safety Information document before operating the vehicle with the product installed.
- This kit is compatible ONLY with the OE or Rekluse clutch components.
- Read this entire document before performing any steps.
- If you install this product for a customer or another person, instruct them to read the **Safety Information** document and the **Installation Guide** before operating the bike with the product.
- Protect eyes and skin – wear safety glasses and work gloves.
- Use the torque values listed in the instructions. Otherwise, use the torque specifications found in your OE service manual.
- For optimal clutch performance, Rekluse recommends using fresh, clean oil that meets JASO-MA oil rating requirements. Rekluse offers **Factory Formulated Oil™** developed specifically for Rekluse products. Rekluse Factory Formulated Oil is a perfect complement to any OEM or aftermarket wet clutch. Visit www.rekluse.com to learn more.

PREPARE BIKE FOR INSTALLATION

1. Stand the bike up on a lift or suitable bike stand.

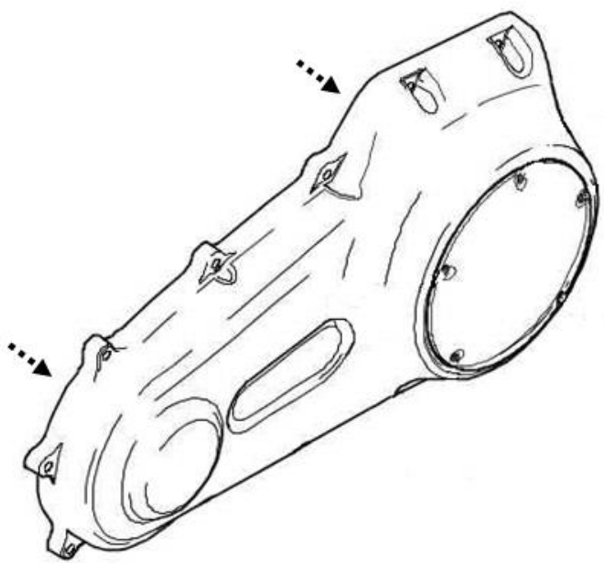
2. Remove the oil drain plug and drain the oil into a suitable container.



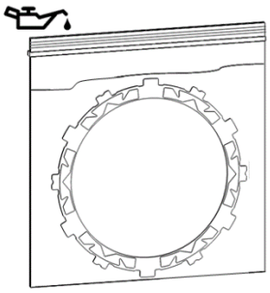
3. Remove any parts that are attached or blocking the primary chaincase cover. These may include the left floorboard, foot peg(s), shift lever, and/or the side stand.

   Note: *Before removing the shift lever, shift the bike into 5th gear.*

4. Remove the primary chaincase cover.



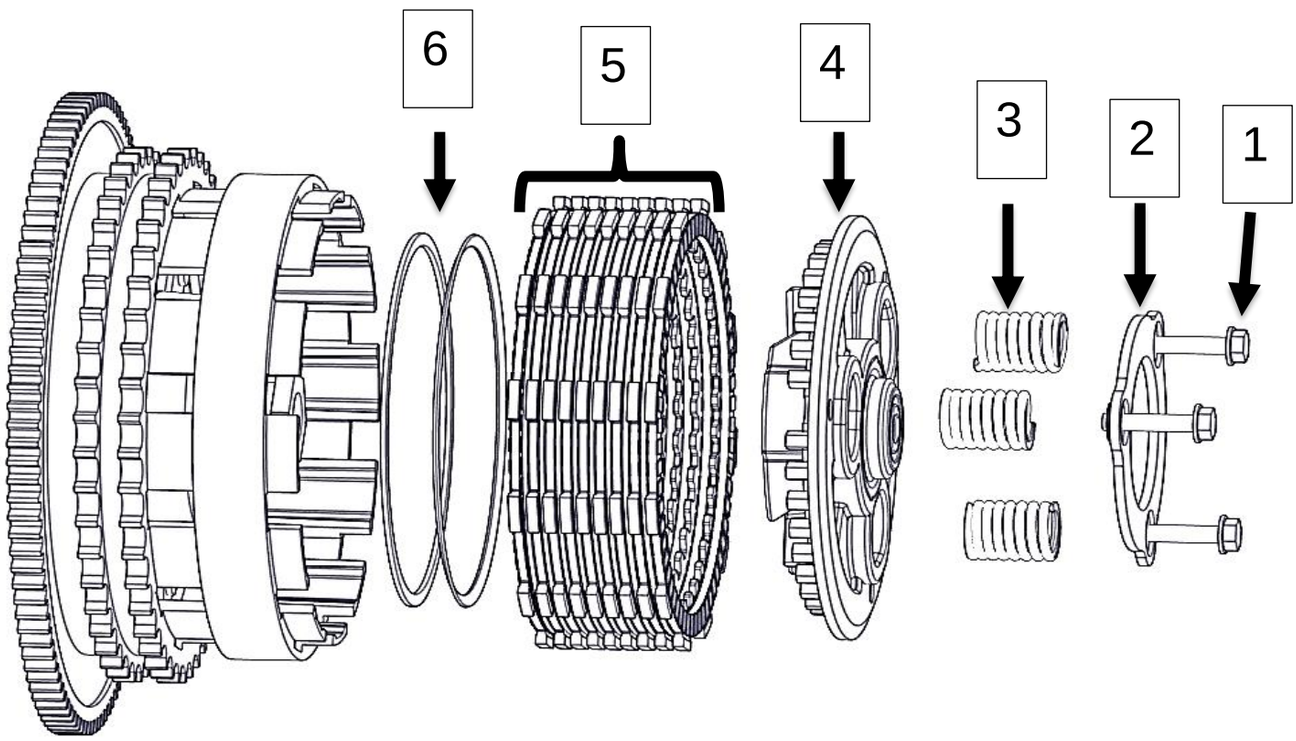
DISASSEMBLE CLUTCH



1. Soak the TorqDrive® friction disks in new primary chaincase oil for 5 minutes. Make sure the friction disks are coated on both sides.

2. Remove the following OE parts. *You may need to use dental pick tools to reach and remove the bottom plates and damper (judder) spring/seat.*

Representative image, your clutch may appear different.



1	Pressure plate bolts
2	Spring hold-down ring
3	Pressure plate springs
4	Pressure plate
5	Clutch pack
6	Damper (judder) spring and seat

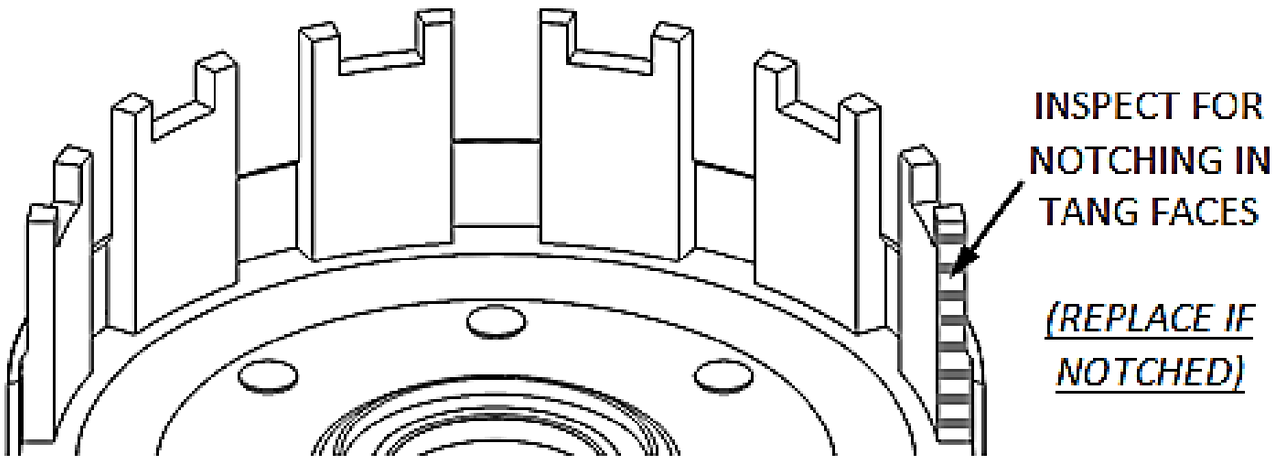
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INSPECT THE BASKET

⚠WARNING

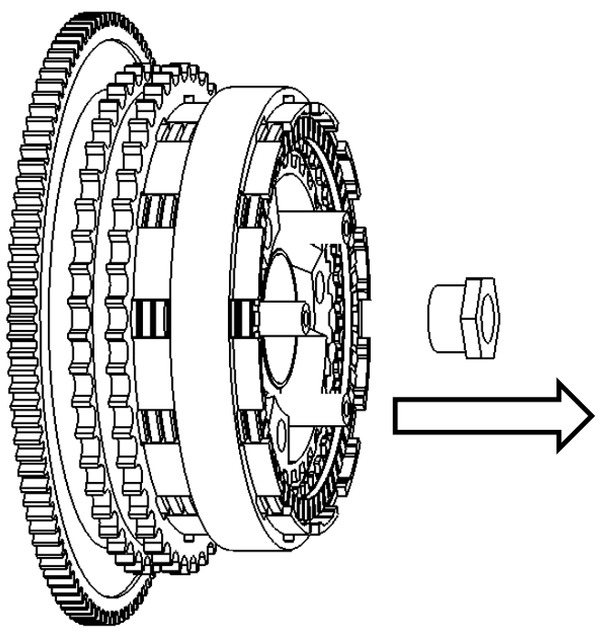
Failure to inspect the basket and replace it if necessary could result in death, serious injury, and/or property damage.

Inspect the clutch basket for notching. Do not install sleeves or use products with a notched basket. Notched basket tang faces can cause the sleeves to break. Do not use baskets that have been filed, machined, or modified on the tangs. Replace the basket if necessary.



REMOVE THE CLUTCH HUB

1. Remove the **left-hand thread** center clutch nut



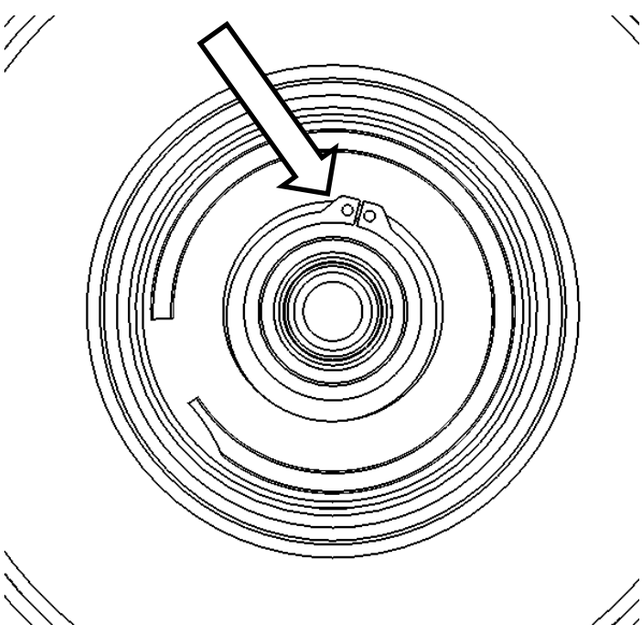
2. Remove the right-hand thread compensator bolt

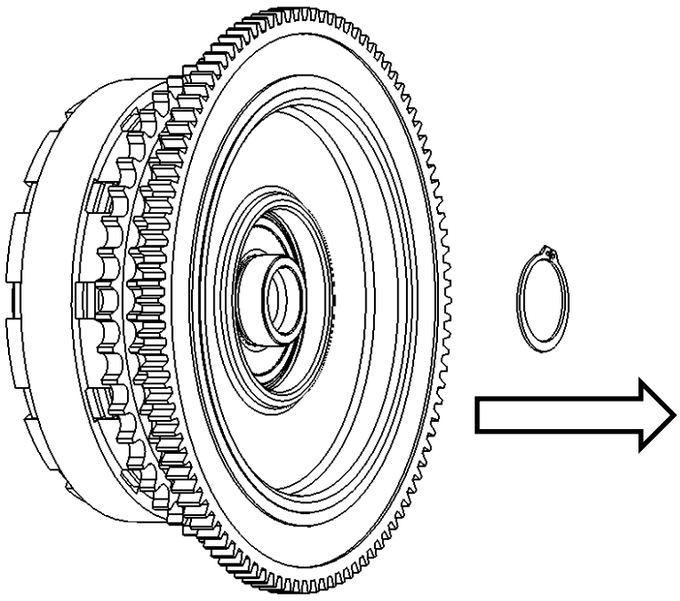
Note: *A brace between the compensator and clutch basket sprockets will be required to keep them from spinning.*

3. Remove the chain tensioner

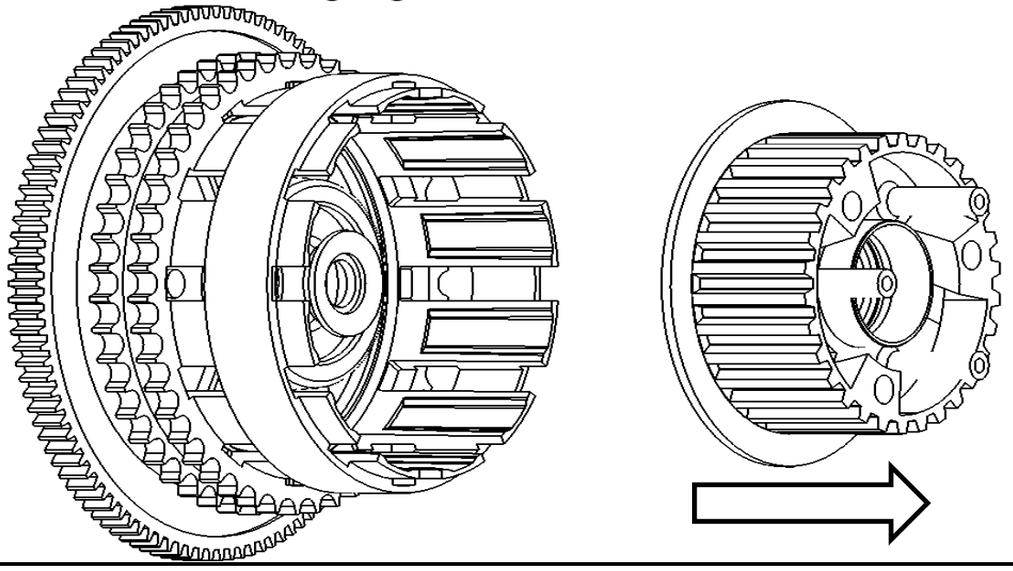
4. Remove the basket & hub assembly along with the primary chain and the compensator sprocket

5. Flip the basket assembly over and remove the smaller snap ring from the hub shaft



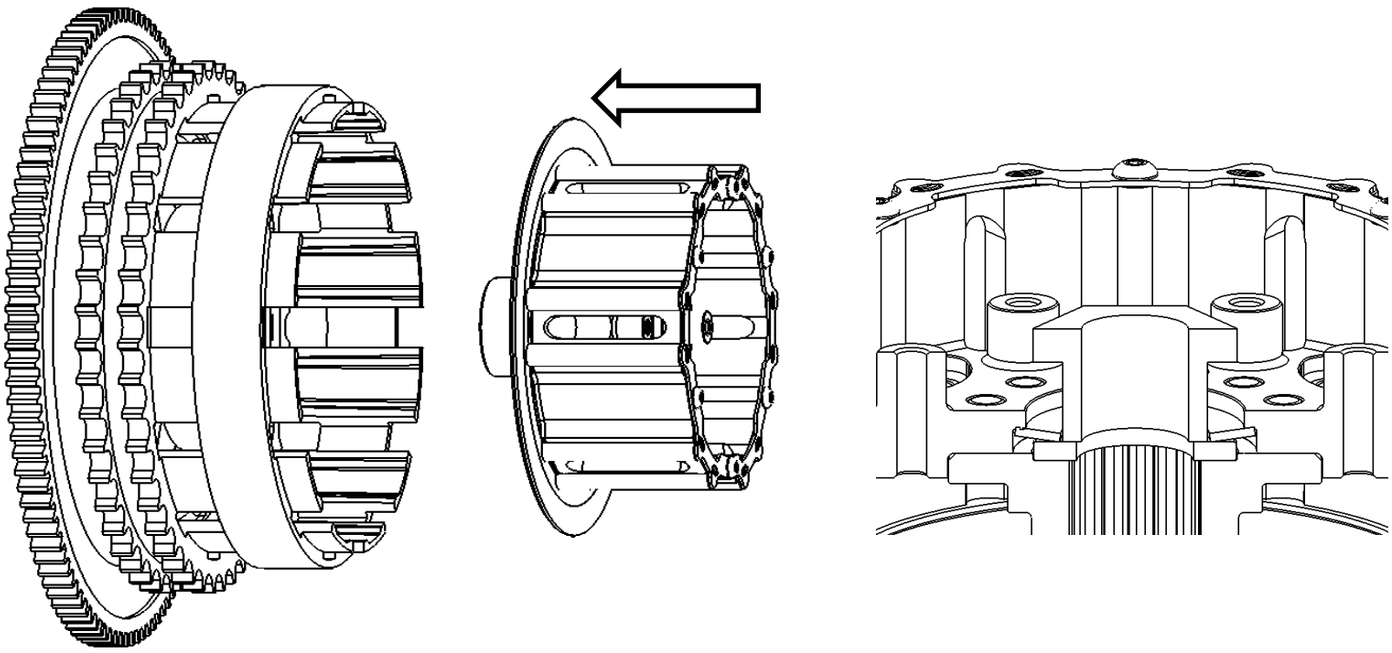


6. Using a hydraulic press, carefully press out the hub from the basket. Support the basket with something soft, such as wood to avoid damaging the basket.



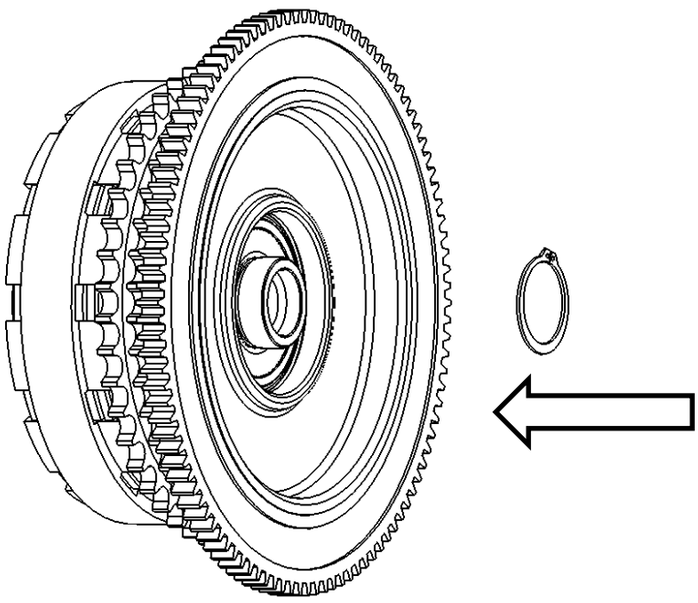
INSTALL THE REKLUSE HUB

1. Press the Rekluse hub into the basket. Do not press directly on the aluminum hub, only on the steel washer in the center.



Note: *It can be helpful to set the hub nut onto the hub when pressing it to give the press something to contact.*

2. Install the OE snap ring onto the hub shaft



3. Reinstall the clutch hub & basket assembly onto the shaft along with the primary chain and the compensator sprocket

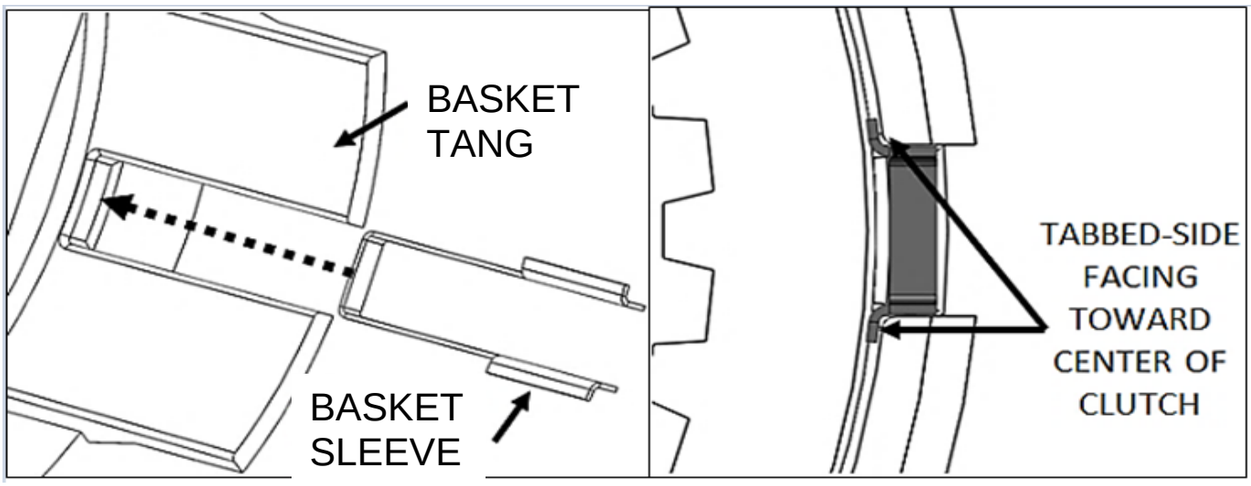
4. Apply two drops of the supplied Loctite to the compensator bolt and torque to **100 ft-lbs**, then back it off ½ turn and torque to **160 ft-lbs** for TwinCam models or **170 ft-lbs** for M8 models**.**

5. Apply two drops of the supplied Loctite to the **Left-hand thread** center hub nut and torque to **70-80 ft-lbs.**

6. Install the primary chain tensioner and torque to **21-24 ft-lbs**.

INSTALL THE BASKET SLEEVES

Install **ALL** the Rekluse basket sleeves into the OE basket slots. Make sure the sleeve tabs sit against the inside of the basket, then push the sleeves down until they contact the bottom of the tang slot.



⚠WARNING

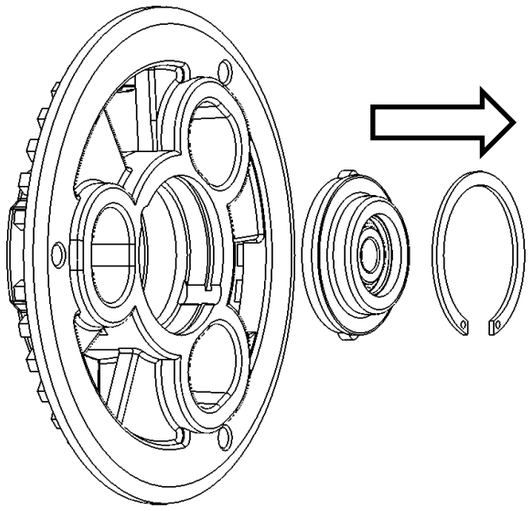
Rekluse basket sleeves are designed to be installed into an OE or Rekluse clutch basket ONLY. The use of non-Rekluse aftermarket clutch baskets may cause clutch damage or failure.

INSTALL THE CLUTCH PACK

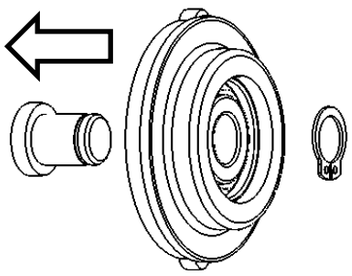
Refer to the included **Setup Sheet** for a breakdown of the clutch pack.

INSTALL THE PRESSURE PLATE

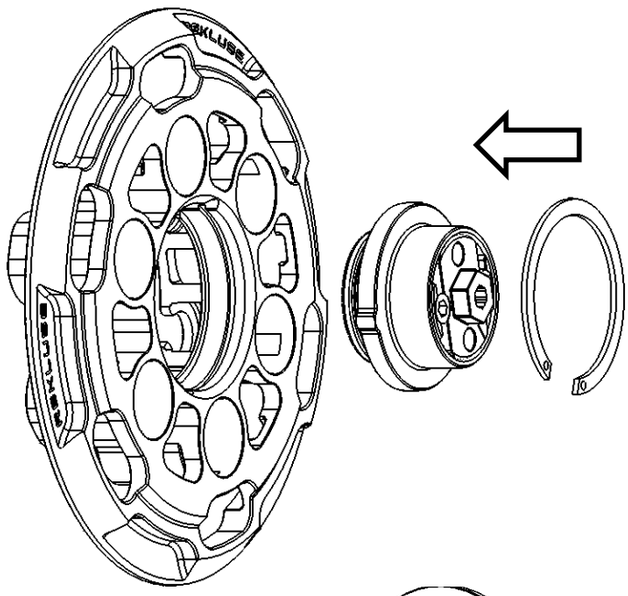
1. Remove the snap ring and bearing from the OE pressure plate.



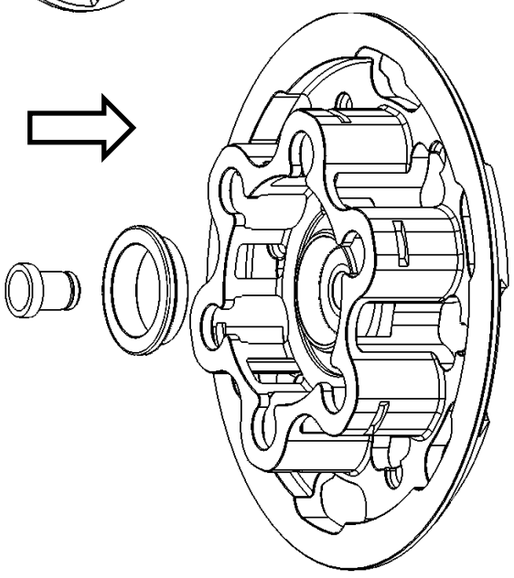
2. Remove the inner snapring and throwout piece from the bearing.



3. Install the included adjuster assembly into the Rekluse pressure plate with the OE snapring. Ensure that the snapring seats properly in the groove.



4. Turn the Pressure plate over. Apply assembly grease to the included steel collar and the OE throwout piece and install them into the bottom of the adjuster assembly on the pressure plate.



INSTALL THE CLUTCH SPRINGS

Select the desired springs from the tables below.

⚠CAUTION

Trikes require the heavy, blue springs. Use of light springs in a trike may result in excessive clutch wear or failure.

Milwaukee 8 Models (17+)

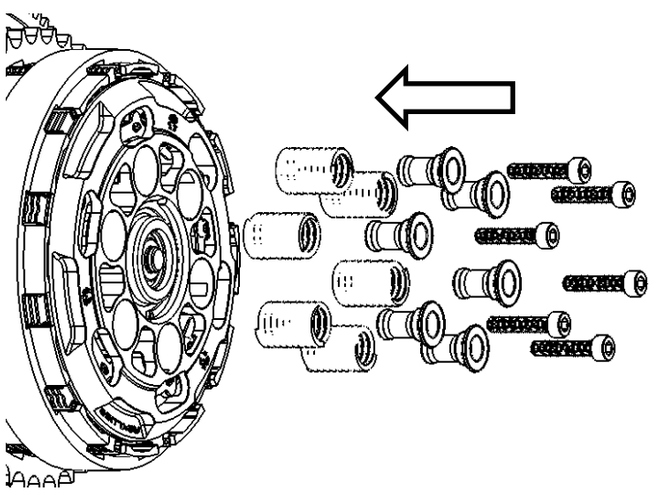
Note: *The OE clutch is rated to hold approximately 125 ft-lbs of engine torque.*

Torque Capacity (lb.ft)	Change in Lever Pull	Color
125	0%	Gold
145	16%	Gold and Blue
170	32%	Blue

Twin Cam Models (07-16)

Torque Capacity (lb.ft)	Change in Lever Pull	Color
125	-18%	Gold
145	-5%	Gold and Blue
170	9%	Blue

1. Install the Rekluse pressure plate, springs, screw sleeves, and screws. Be sure to alternate springs if mixing colors.
2. Torque the pressure plate screws to **9 lb-ft (12 N-m)**.



SET THE INSTALLED GAP

The “installed gap” is the gap set in the clutch pack that allows the auto-clutch to disengage and engage automatically.

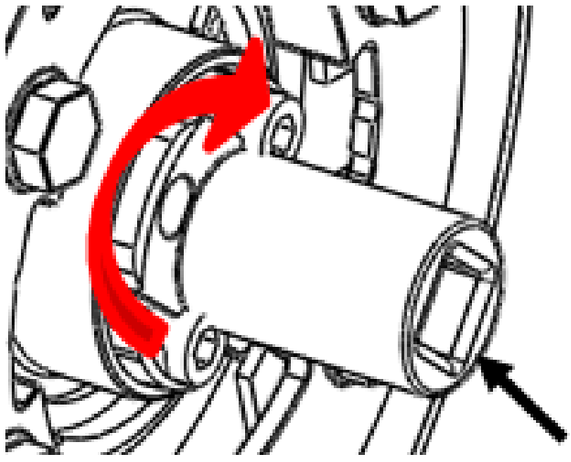
⚠WARNING

Failure to set and maintain the installed gap can result in clutch failure. The gap must be set properly for the clutch to function

1. Remove the cap from the master cylinder reservoir. Be sure to protect the bike during the adjustment process by catching any fluid that might overflow into a suitable container.

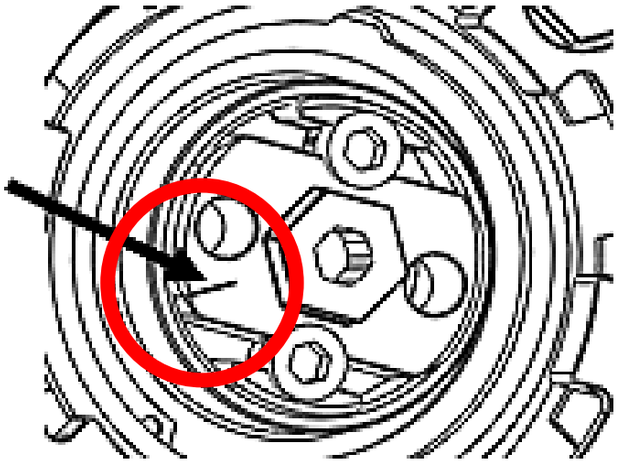
2. Turn the adjuster screw **CLOCKWISE** until it stops under moderate pressure.

   - You are feeling for the point at which it bottoms out and starts to lift the pressure plate (you will feel an abrupt increase in turning effort). This position is called your **starting point**.



Use 1/2” socket or 7/32” hex key

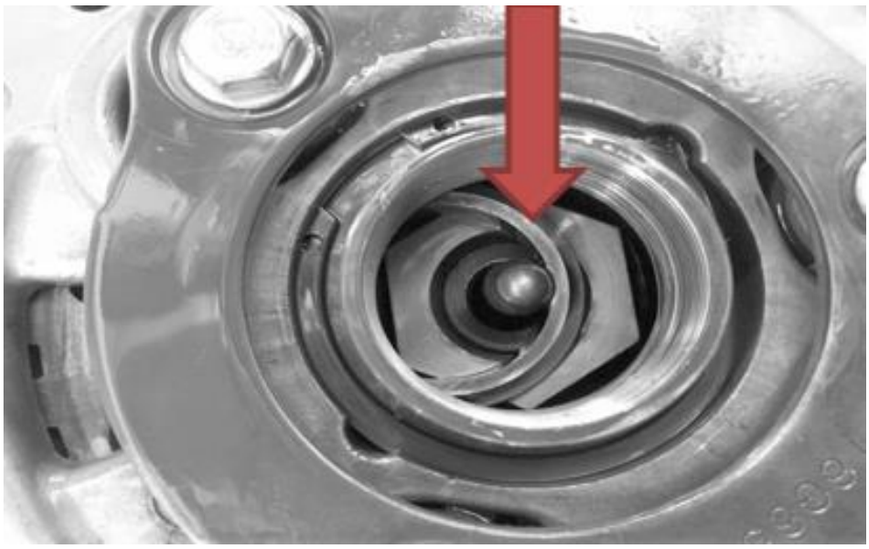
3. At this starting point, note the location of the tick mark on the aluminum adjuster. Marking the tick mark with a marker can help make it more visible during the adjustment.



4. Turn the adjuster **CLOCKWISE** 1+3/8 turns from the starting point.

Notes:

- *If the adjuster is not turning smoothly, then the adjuster collar may be unindexed from the adjuster assembly. Uninstall the adjuster assembly and reindex the adjuster collar.*



- *After setting the installed gap, the aluminum center adjuster should be flush or near flush in the adjuster assembly.*



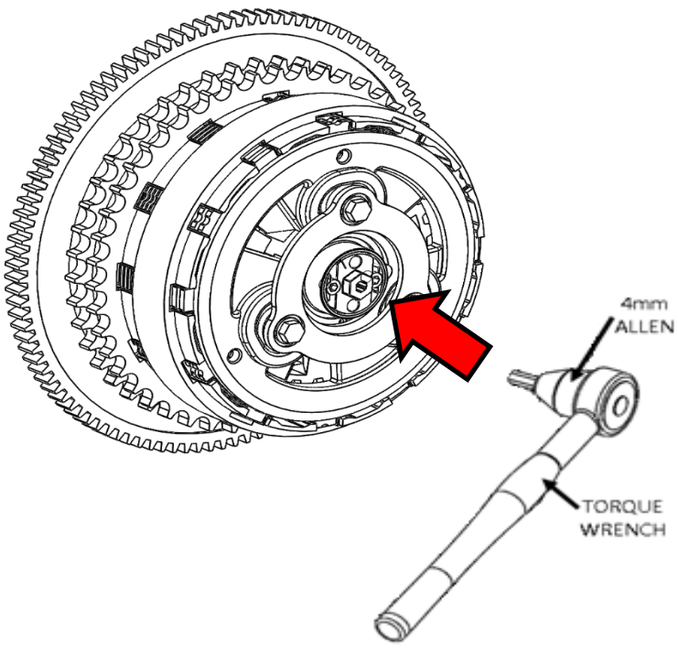
- *If the adjuster is deeper than flush, a clutch plate may be unindexed from the hub.*



- *If the adjuster is protruding out of the adjuster assembly the adjuster collar likely has unindexed while installing the pressure plate.*



5. Use a 4 mm hex socket and torque wrench to lock the adjuster in place. Alternate tightening the two pipe plugs in small increments to **40 in-lb (4.5 N-m).**



Note: When torqued correctly, the tops of the plugs should be nearly flush with the top of the aluminum adjuster piece.

Warning Sticker

Install the provided warning sticker on the backside of the clutch lever, such that it is visible to the rider.

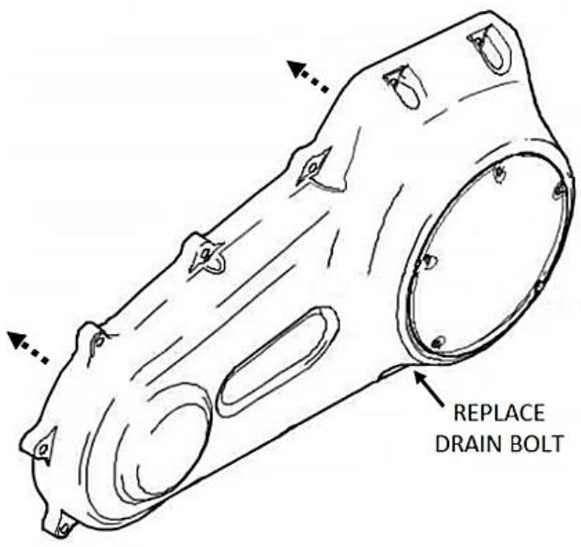


INSTALL THE PRIMARY COVER

1. Thoroughly clean the mating surfaces of the primary cover and the engine case.

2. Reinstall the primary gasket (or new gasket).

3. Reinstall the primary cover, then torque the cover bolts to **84-108 in-lb (9.5-12.2 N-m)**.

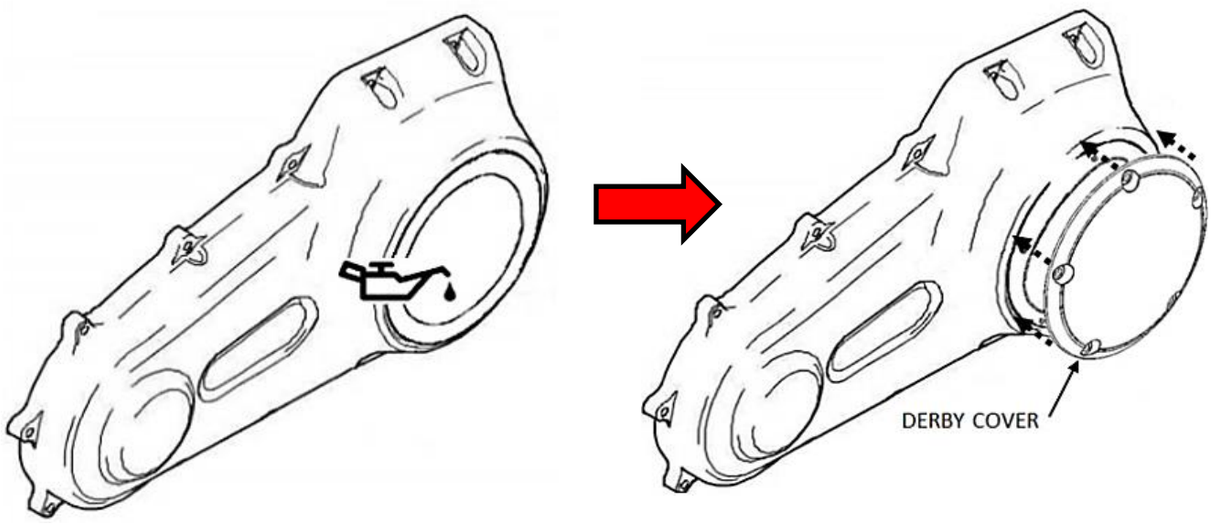


4. Reinstall the drain bolt, then torque the drain bolt to **36-60 in-lb (4-6.8 N-m)**.

5. With the primary cover installed, use a T-27 Torx bit to remove the derby cover.

6. Using a funnel, add the OE recommended oil type and quantity (typically **1 liter**) to the primary case through the derby cover cavity. Rekluse recommends our **Factory Formulated Oil.**

7. Reinstall the derby cover and torque bolts to **84-108 in-lbs. (9.5-12.2 N-m).**



FREE PLAY GAIN

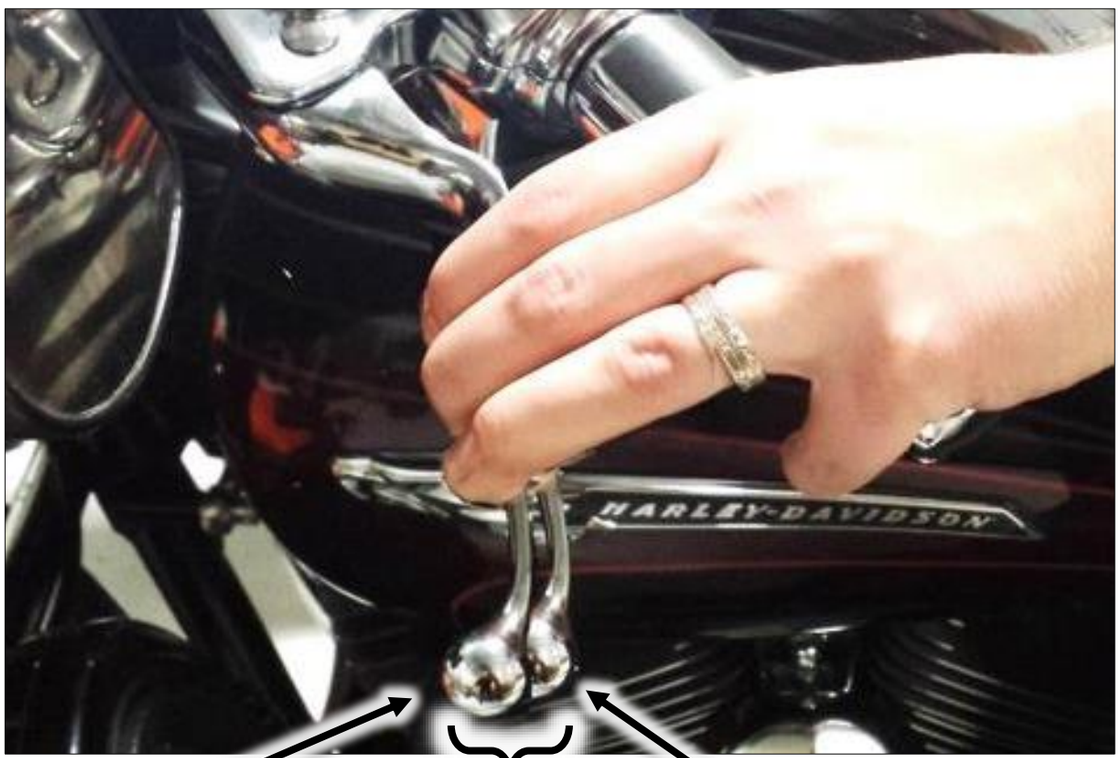
Free Play Gain is the increased free play movement in the clutch lever when the throttle is blipped in Neutral. Free play gain allows the installed gap to be checked quickly and without opening up the bike.

Correct Free Play Gain = Correct installed gap

Maintaining the installed gap is CRUCIAL. Use the following steps to verify the installed gap by checking Free Play Gain.

⚠WARNING

Failure to check and verify Free Play Gain can cause failure or damage to this product. Setting the correct installed gap is critical for clutch performance.



Lever with “slack” removed

Lever position around 4,000 RPM

Free Play Gain
1/8”-1/4” (3 mm-6 mm) lever movement

- **If there is too much Free Play Gain,** the installed gap is too small.
  - The bike may drag and stall because it has difficulty disengaging the clutch. It may also be difficult to shift.
  - Too much Free Play Gain will not hurt the clutch, but it will negatively affect clutch performance.

- **With too little or no Free Play Gain**, the installed gap is too large.
  - The clutch may slip and make the bike seem like it is losing power
  - The bike may not move forward even though the engine RPM increases
  - Too little Free Play Gain will cause the clutch system to burn up

Optimal Free Play Gain yields 1/8”-1/4” (3 mm-6 mm) of clutch lever movement, measured at the ball end of the lever. This measurement at the lever correlates to achieving the ideal installed gap.

A video showing how to check Free Play Gain is also available at: https://rekluse.com/support/videos.

⚠WARNING

BEFORE YOU BEGIN, verify that the bike is in NEUTRAL before checking Free Play Gain. Failure to do so may result in the bike lurching forward, and loss of control and/or injury may result.
A Rekluse auto-clutch can make your motorcycle appear to be in neutral when in gear, even when the engine is running and the clutch lever is released.
Motorcycles equipped with a Rekluse auto-clutch can move suddenly and unexpectedly and cause riders to lose control. To avoid death, serious injury, and/or property damage, always sit on the motorcycle to start it.

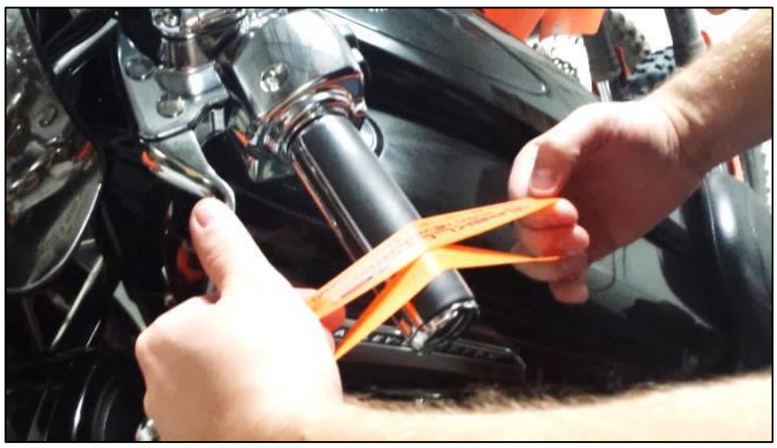
Two Ways to Check Free Play Gain

1 - The Rubber Band Method

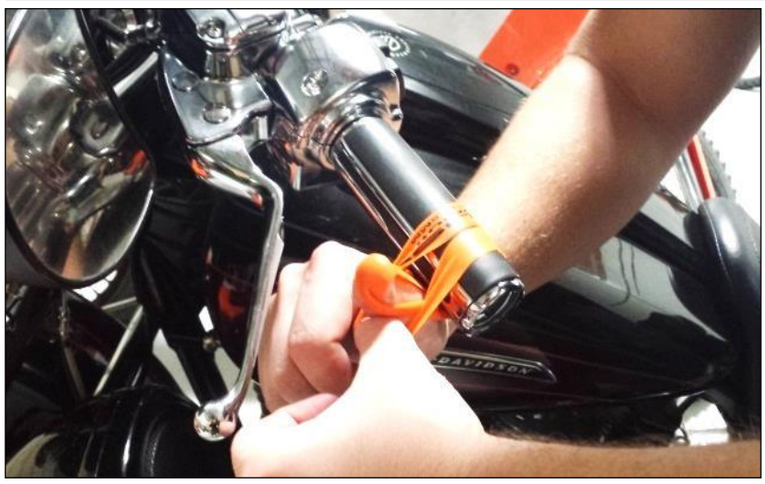
This is the recommended way to learn how to check. Use the rubber band method for the initial setup. It can also be used before each ride until you feel comfortable checking the Free Play Gain using the hand method.

a) Before you begin, place the bike in **NEUTRAL**, start the engine, and let it warm up.

b) Stretch the included rubber band between your thumbs, then place the top end of the rubber band on the outer end of the left handlebar grip.



c) While holding the top end of the rubber band against the handlebar, stretch the band downward, then loop it through itself.



d) Pull the band through the loop, then attach it to the outside end of the clutch lever. This will take up the initial free play (slack) and put the lever in a position to detect the Free Play Gain.



e) While still in **NEUTRAL**, quickly rev the engine between 3,000-5,000 RPM, then let it return to idle. Notice the movement in the clutch lever when the engine is revved. This is your Free Play Gain.

Note: *It is very important the motor returns to idle before revving the engine again or Free Play Gain will not be correct.*

f) When the bike returns to idle, rest your hand across the clutch lever. Rev the engine again to 3,000-5,000 RPM so you can observe the movement while feeling Free Play Gain with your hand.

2 - The Hand Method

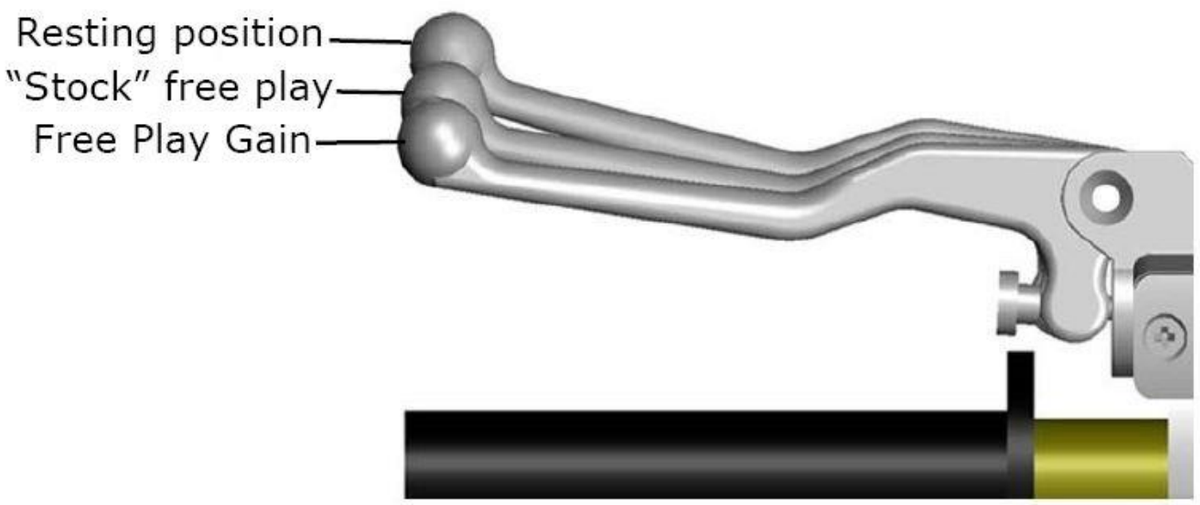
Use the hand method to check Free Play Gain before the start of every ride for optimum performance and longevity of your new clutch.

a) Before you begin, place the bike in **NEUTRAL**, start the engine, and let it warm up for 2-3 minutes to idle down and warm up the engine oil.

b) With the bike at idle, apply enough pressure to the clutch lever to take up the initial free play (slack) in the clutch lever.

c) While still in **NEUTRAL**, continue to apply light pressure and quickly rev the engine between 3,000-5,000 RPM, then let it return to idle. Notice the movement in the clutch lever when the engine is revved. This is your Free Play Gain.

d) When the bike returns to idle, rev the engine between 3,000-5,000 RPM a second time to verify the Free Play Gain again.



ADJUST THE INSTALLED GAP

After checking for Free Play Gain, you may need to adjust the installed gap. If Free Play Gain is optimal, continue to “BREAK IN THE NEW CLUTCH.” If Free Play Gain is not optimal, the installed gap needs to be adjusted.

The installed gap should be fine-tuned in small increments and then recheck Free Play Gain. Refer to the table below to set the properly installed gap based on your Free Play Gain.

Symptom	Reason	Solution
• Too much Free Play Gain: The clutch lever moves too far • Clutch has excessive drag or stalls • It is difficult to fully override the clutch with the lever	The installed gap is too small	Turn the adjuster **clockwise** to increase the Installed Gap. Recheck Free Play Gain.
• Too little Free Play Gain: The clutch lever only moves slightly or does not move at all • Clutch slips • The bike seems to have lost power	The installed gap is too large	Turn the adjuster **counterclockwise** to reduce the Installed Gap. It may be helpful to re-find the starting point. Recheck Free Play Gain.

BREAK IN THE NEW CLUTCH

New clutch components must be broken in before use. Follow this procedure any time new clutch components are installed.

Note: *If the engine wants to stall or the creep is excessive, the idle may be too high or the installed gap may be too small. Make necessary adjustments before proceeding.*

Note: *Your clutch pack will expand with heat, so the final adjustment to Free Play Gain should be made when the bike is warm. Remember not to ride without sufficient Free Play Gain.*

⚠CAUTION

Never start in a gear other than first. Always keep the motorcycle in first gear when taking off from a stop. Taking off from a higher gear can cause premature clutch wear and damage to the product.

DO NOT DYNO TEST YOUR MOTORCYCLE BEFORE BREAK-IN! Always break in the product before performing dyno testing. Read the included dynamometer sheet for more information.

⚠WARNING

Failure to follow the break-in procedure could cause motor oil delivery failure, which can result in motor failure, serious injury, or death.

Break-in Procedure	Number of times
1. Warm up the bike for 2-3 minutes. With the bike in **NEUTRAL** and your hand **off** of the clutch lever, rev the engine 10 times, being sure to let it **return to idle** between each rev cycle.	
2. With the engine still running, pull in the clutch lever, then shift the bike into 1st gear. Slowly release the clutch lever. The bike should stay running and in place, or have a slight amount of forward creep. 3. With the bike idling in first gear, slowly apply the throttle to begin moving. 4. Without using the clutch lever, accelerate moderately to approximately 3,500 RPM to fully lock up the clutch and come to a complete stop. Repeat 15 times.	 15 roll-on starts
5. Place the bike in **NEUTRAL** and recheck Free Play Gain. Continue to adjust the installed gap until the clutch lever is 1/8”-1/4” (3 mm-6 mm).	 Recheck Free Play Gain and adjust

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Doc Rev: 091224

EXP TUNING OPTIONS

The EXP can be tuned to engage at a higher or lower RPM. The Standard setting is pre-installed in the EXP.

Twin Cam Spring Settings

Idle RPM	Engagement Springs	Engagement setting
< 850 RPM	3 Green, 3 Gold	Low
850 - 1050 RPM	6 Green	Standard
1050 - 1150 RPM	3 Green, 3 White	High
> 1150 RPM	6 White	Extra-High

M8 Spring settings

Idle RPM	Engagement Springs	Engagement setting
< 850 RPM	6 Green	Low
850 - 1050 RPM	3 Green, 3 White	Standard
1050 - 1150 RPM	6 White	High

EXP setting is CORRECT:

- The clutch engages smoothly and right above idle
- Some cold drag may be present but is gone when warm

EXP setting too HIGH:

- Engagement is delayed and feels jerky
- Bike freewheels easily

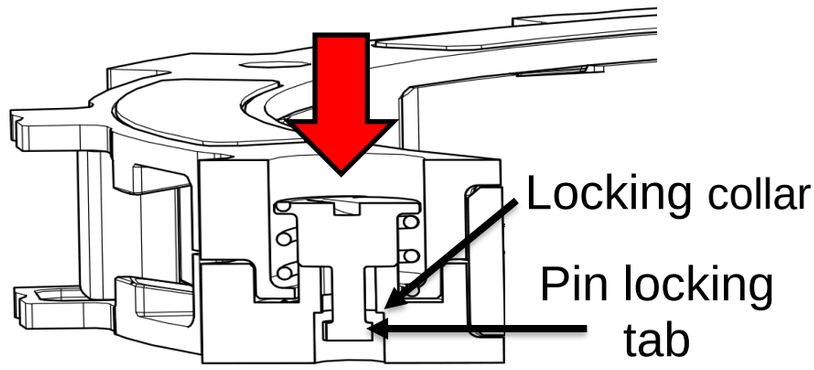
EXP setting too LOW:

- Clutch drags, even with proper free play gain and warmed-up
- The engine feels bogged down at low RPM

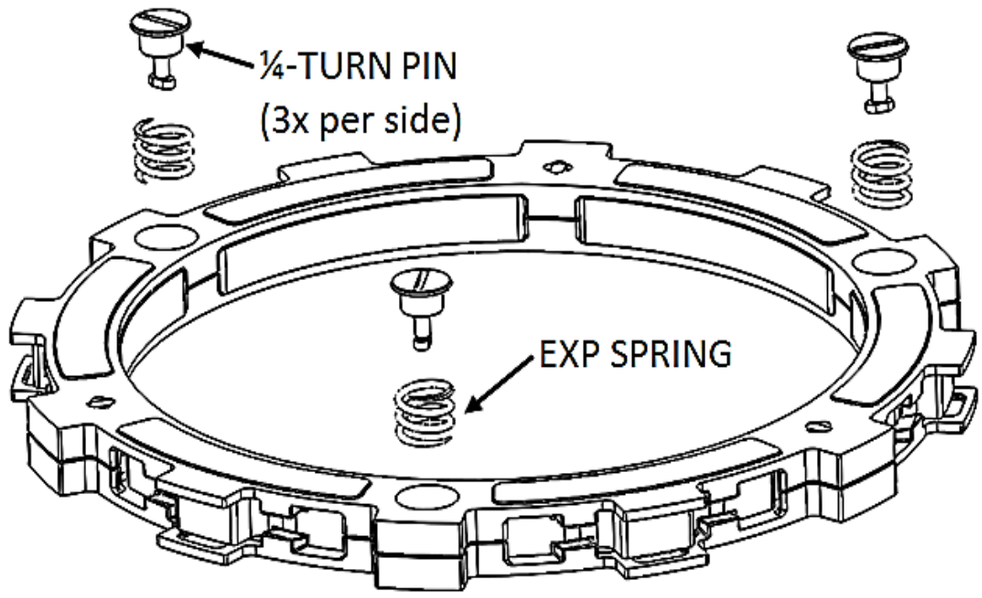
Note: *Modified engines may require the high or extra high setting.*

Changing the springs

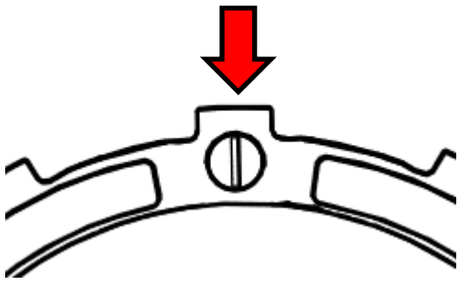
1. Using a flat-head screwdriver, push the ¼ turn pin all the way in and gently turn it 90° to release it.



2. Replace springs one side at a time. This prevents having to disassemble the EXP completely.



NOTE: *Lining up the slot in the pin with the tab in the EXP will align the pin in the proper position for locking the pin.*



3. If you need to disassemble the EXP disk, you can watch the video on our website under Tech Tips at www.rekluse.com/support/videos.

Note: *To maintain even pressure, when using two different color spring sets, install one set of 3 on one side of the EXP and the remaining set of 3 on the other side.*

MAINTENANCE

To keep your clutch performing at its best, perform regular maintenance on your bike and clutch.

- Keep up with regular oil changes according to the bike manufacturer’s recommendations. Clutch performance and longevity depend on oil quality. Tired, dirty, or worn oil may cause excessive clutch drag or noise.
- Use oil recommended by the manufacturer of your bike.
- For optimal clutch performance, Rekluse recommends using fresh, clean oil that **meets JASO-MA or MA2** standards.
- Inspect all of your clutch parts for signs of wear or excessive heat, and replace components as necessary. This includes your basket sleeves. Clutch wear is dependent on the rider's use.
- Replace friction disks and EXP if they measure below the specifications found in your setup sheet, or if the disks are glazed and/or burnt.
- Repeat the break-in procedure anytime you replace the friction disks. Always soak friction disks in oil for at least 5 minutes before installing.
- Replace the drive plates if they show signs of excessive heat.

Disk inspection examples

When inspecting the clutch pack, the following pictures can be used as a reference. **These are best viewed in color by viewing this install document on www.rekluse.com/support.**

Drive Plates – If the clutch pack is getting high amounts of heat, purple, blue, or black color can be seen on the drive plate teeth. See the pictures below. Not all drive plates look the same and may look different than pictured.



Normal Heat



High Heat (Blue)



Excessive Heat (Black)

Friction Disks – Due to the dark color of the friction material, the friction disks will appear almost black as soon as they are put in oil. During the inspection, look for glazing of the friction material. Glazing will appear shiny and feel like glass, even after the oil is cleaned from the friction disk. Not all friction disks look the same and may look different than pictured.



Normal Friction



Glazed Friction

TROUBLESHOOTING

Clutch Drag:

- Make sure the bike has reached operating temperature. Drag may occur when the engine is cold
- Check that freeplay gain is set properly. Incorrect freeplay gain may cause drag.
- Check idle speed. If idle speed is too high, drag may occur. Refer to the EXP adjustment chart for further tuning.
- Put in new Rekluse recommended oil. Old or improper oil can cause performance issues.
- Check the clutch pack for signs of excessive heat or warpage.

Clutch Slip:

- Inspect the clutch for signs of excessive heat.
- Check that freeplay gain is set properly. Incorrect freeplay gain can cause slip.
- If clutch slip persists, heavier clutch springs are recommended

LEVER SAFETY STRAPS

Your kit includes 2 Velcro-type straps to be used to secure both the clutch and front brake levers when the bike is parked.

Use the lever safety straps every time you park or leave the bike. **Refer to the Safety Information sheet for more information.**

1. Pull the lever tight against the handlebar.
2. Wrap the Velcro safety strap around the lever and handlebar, pull it tight, and then fasten it.



Clutch Lever Strap: to prevent unwanted launching.



Brake Lever Strap: for use as a parking brake.

NEED ADDITIONAL HELP?

Website

www.rekluse.com/support

Phone

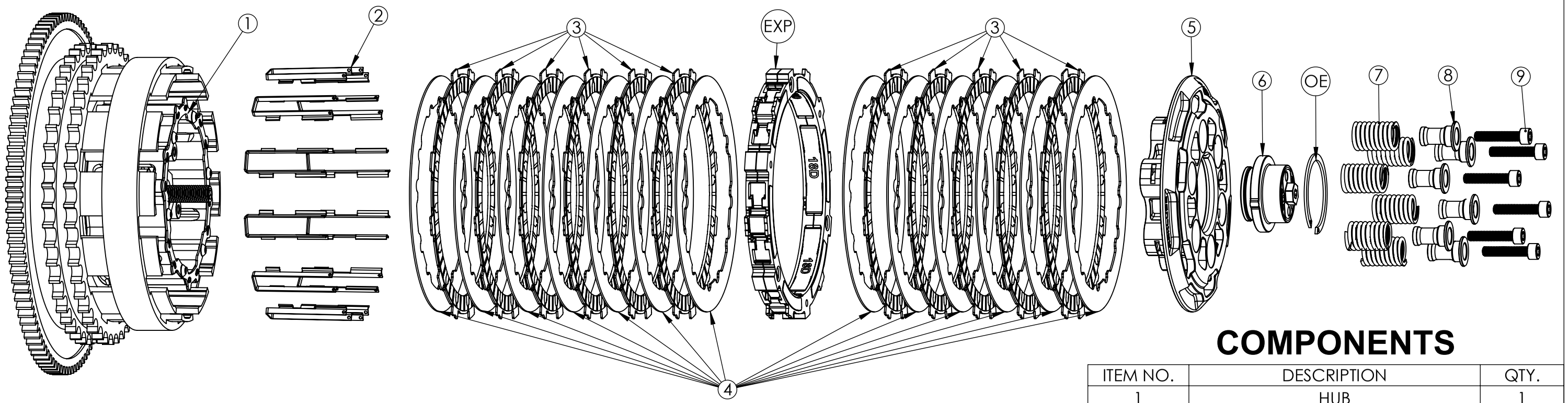
(208) 426-0659

Monday thru Friday: 8 am – 5 pm Mountian Time

Email

customerservice@rekluse.com

SETUP SHEET 198-7915005



COMPONENTS

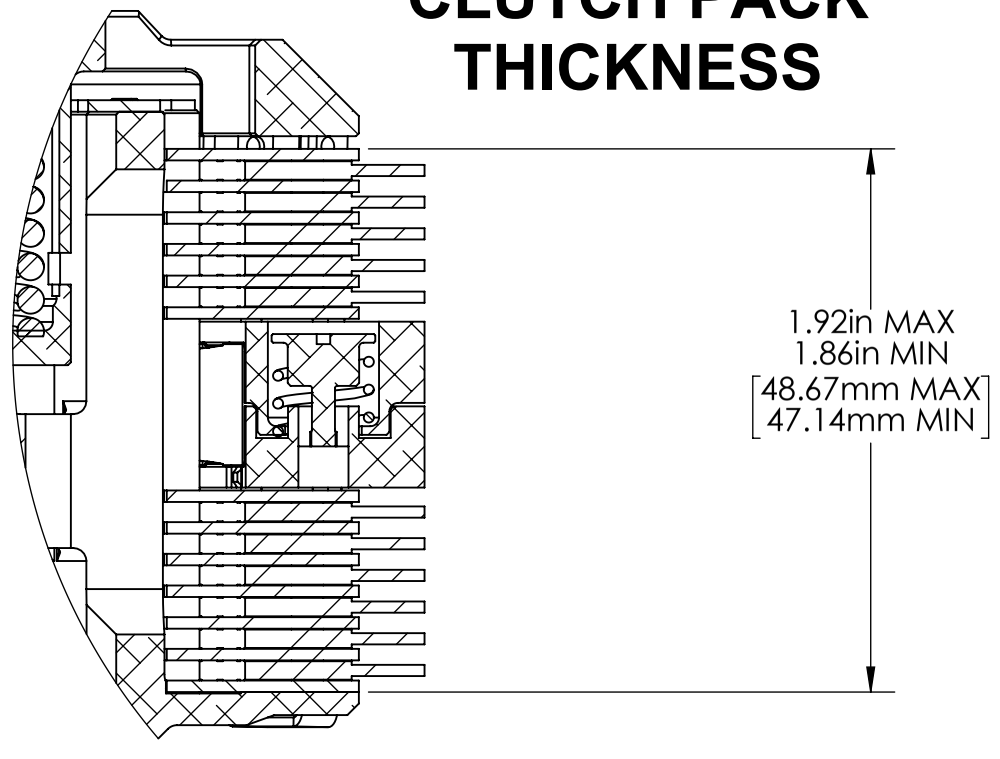
ITEM NO.	DESCRIPTION	QTY.
1	HUB	1
2	BASKET SLEEVES	12
3	TORQ DRIVE FRICTION	11
4	DRIVE PLATE	13
5	PRESSURE PLATE	1
6	THROW-OUT ADJUSTER	1
7	PRESSURE PLATE SPRINGS	6
8	SCREW SLEEVE	6
9	SPRING SCREW	6
EXP	EXP DISC	1
OE	OE COMPONENTS	Var

TUNING OPTIONS

ENGAGEMENT RPM	EXP SPRINGS
LOW	6 GREEN
MEDIUM *	3 GREEN & 3 WHITE
HIGH	6 WHITE

* MEDIUM SETTING IS PREINSTALLED IN THE EXP DISC

CLUTCH PACK THICKNESS



SERVICE LIMITS

COMPONENT	STANDARD	SERVICE LIMIT
TORQDRIVE FRICTION	.068-.072in 1.73-1.83mm	.065in 1.65mm
EXP	.596-.616in 15.1-15.6mm	.586in 14.9mm

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